| Doc Number: 161310G00000-00 | Doc Version: 008 | Date: 02-Nov-2026 |
|---|---|---|

# Super Mushshak Automatic Flight Test

Prepar3D v5 | Rahwali Airfield (OPRW) | Automatic Testing Mode

**Super Mushshak Logger - ver. 1.0**

Super Mushshak Airborne Automatic Testing Suite

Flight Test File: Super_Mushak_auto_takeoff_data_2026-11-02_14-56-53.csv

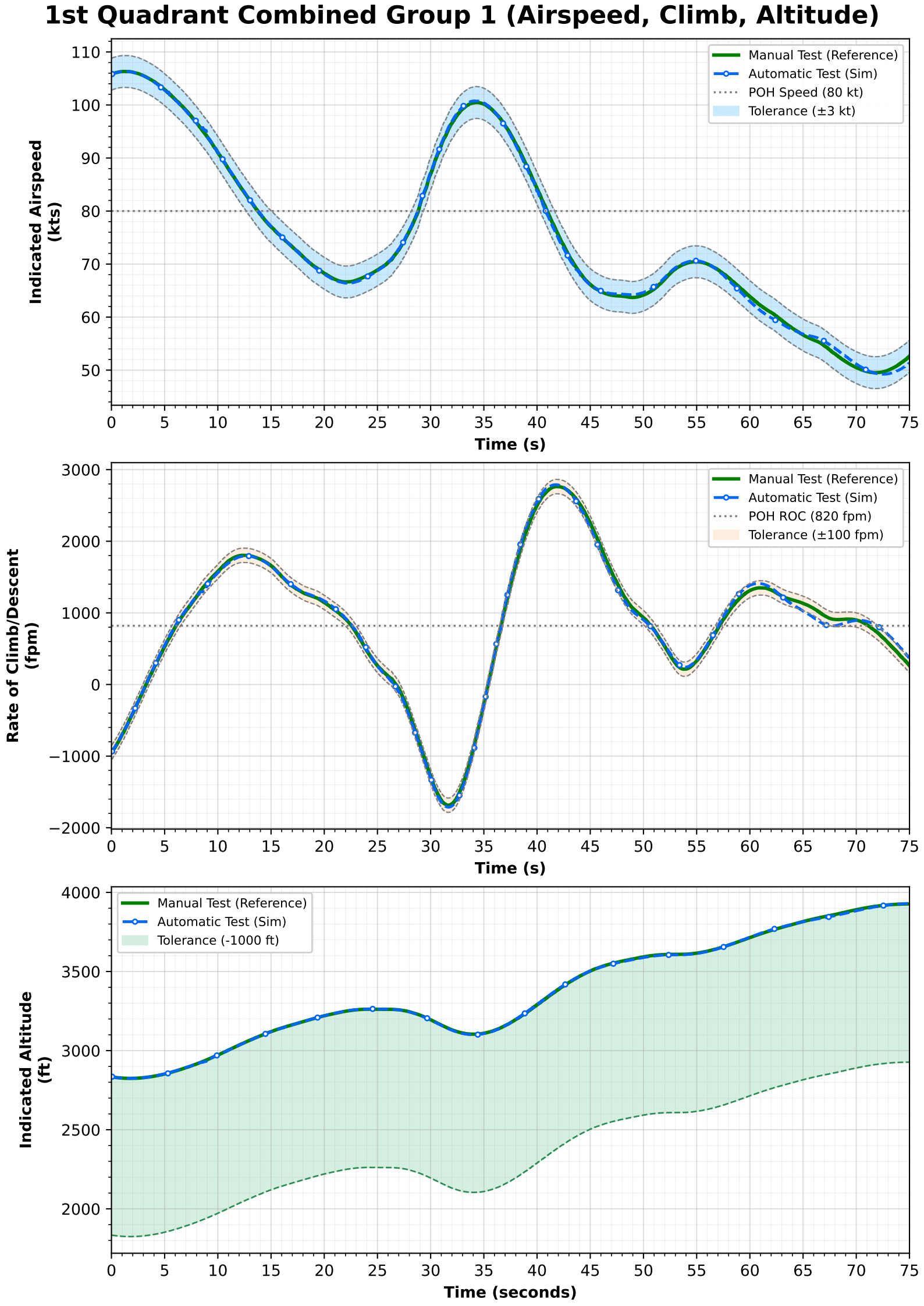
**ONLINE, AUTOMATIC**

Test Date: 02-Nov-2026  Test Time: 14:56:53

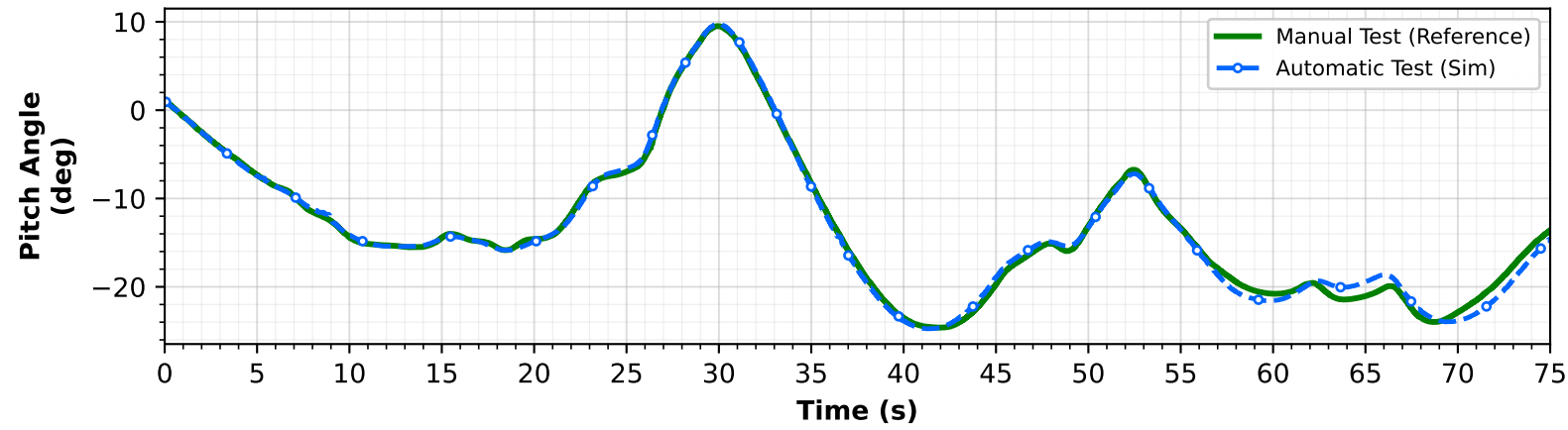
## Initial Test Conditions

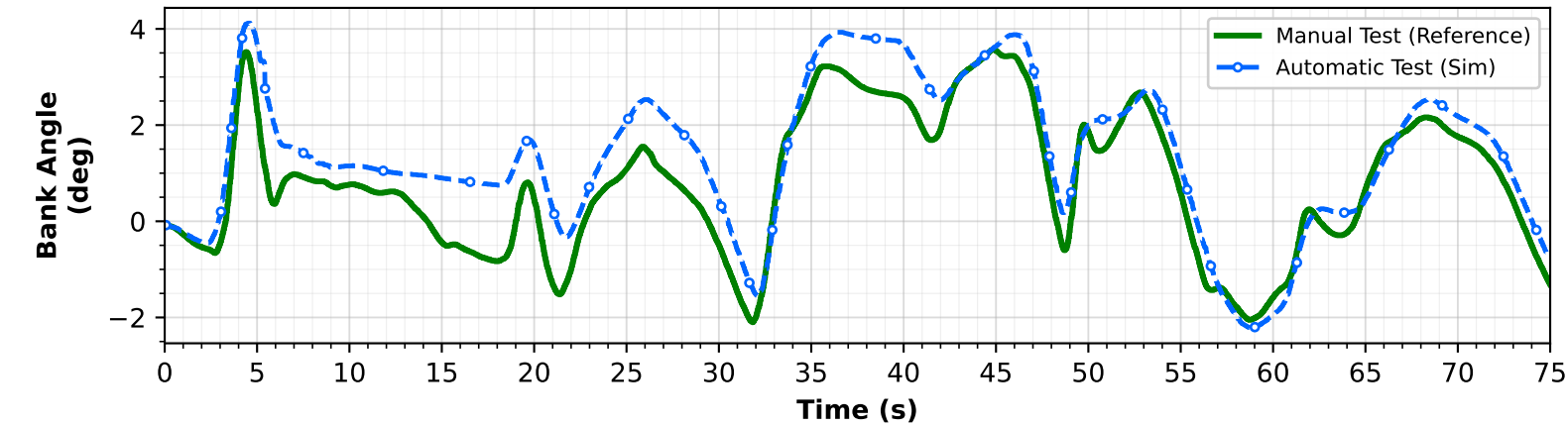
| Validation | Sim | |
|---|---|---|
| ***** Weight and Balance ***** | | |
| 845.5 | 845.5 | Gross Wt (kg) |
| 106.4 | 106.4 | Total Fuel Quantity (kg) |
| 0.00 | 0.00 | CG Pos in X Axis (mm) |
| 0.00 | 0.00 | CG Pos in Y Axis (mm) |
| ***** Control Parameters ***** | | |
| -2.54 | -2.58 | Pitch Control Position (raw) |
| -0.47 | -0.33 | Roll Control Position (raw) |
| 0.00 | 0.00 | Rudder Pedal Position (raw) |
| 81.4 | 81.4 | Throttle Position (%) |
| -2.18 | -1.12 | Pitch Force: Applied/Virtual (N) |
| 0.00 | -0.00 | Roll Force: Applied/Virtual (N) |
| 0.00 | -0.00 | Yaw Force: Applied/Virtual (N) |
| ***** Environment Parameters ***** | | |
| 1013 | 1013 | Baro Press (hPa) |
| 10.9 | 10.9 | OAT (deg C) |
| 325.2 | 325.2 | Wind Dir (mag,deg) |
| 7.70 | 7.70 | Wind Spd (kt) |
| ***** Engine Parameters (Piston Engine) ***** | | |
| 2700 | 2693 | Engine Speed (RPM) |
| 81.4 | 81.4 | Throttle Position (%) |
| 0.00 | 0.00 | Fuel Flow (pph) |
| 534.6 | 533.5 | Exhaust Gas Temp (deg C) |

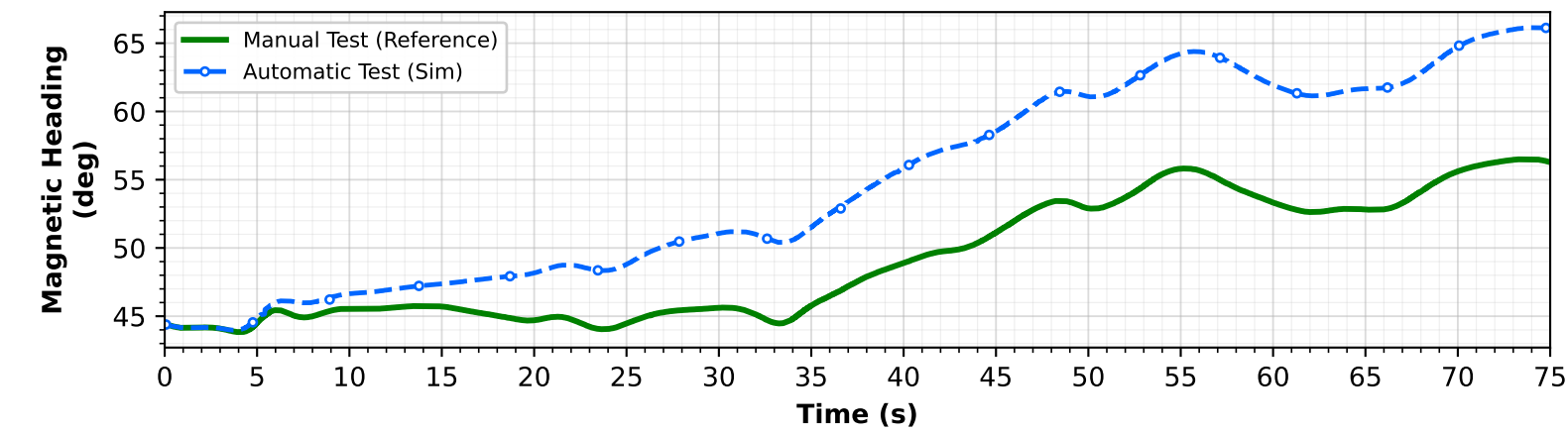
| Validation | Sim | |
|---|---|---|
| ***** Flight Parameters ***** | | |
| 105.8 | 105.8 | Indicated Airspeed (kt) |
| 109.0 | 109.1 | Ground Speed (kt) |
| 0.17 | 0.17 | Mach Number (-) |
| 2832 | 2832 | Pressure Alt (ft) |
| 2087 | 2087 | Radar Alt (ft) |
| 750.0 | 750.0 | Height of Terrain (ft) |
| -943.5 | -933.5 | Vertical Velocity (ft/min) |
| 1.01 | 0.94 | Pitch Angle (deg) |
| -0.08 | -0.08 | Bank Angle (deg) |
| 44.4 | 44.4 | Heading (deg) |
| 1.88 | 1.88 | Aircraft Angle of Attack (deg) |
| -0.11 | -0.04 | Sideslip Angle (deg) |
| 48.3 | 44.5 | Track Angle (deg) |
| 0.08 | 0.08 | Roll Rate (deg/s) |
| -1.72 | -1.73 | Pitch Rate (deg/s) |
| -0.38 | -0.37 | Yaw Rate (deg/s) |
| 1.20 | 1.14 | X Body Acceleration (g) |
| 0.05 | 0.04 | Y Body Acceleration (g) |
| 1.23 | 1.44 | Z Body Acceleration (g) |

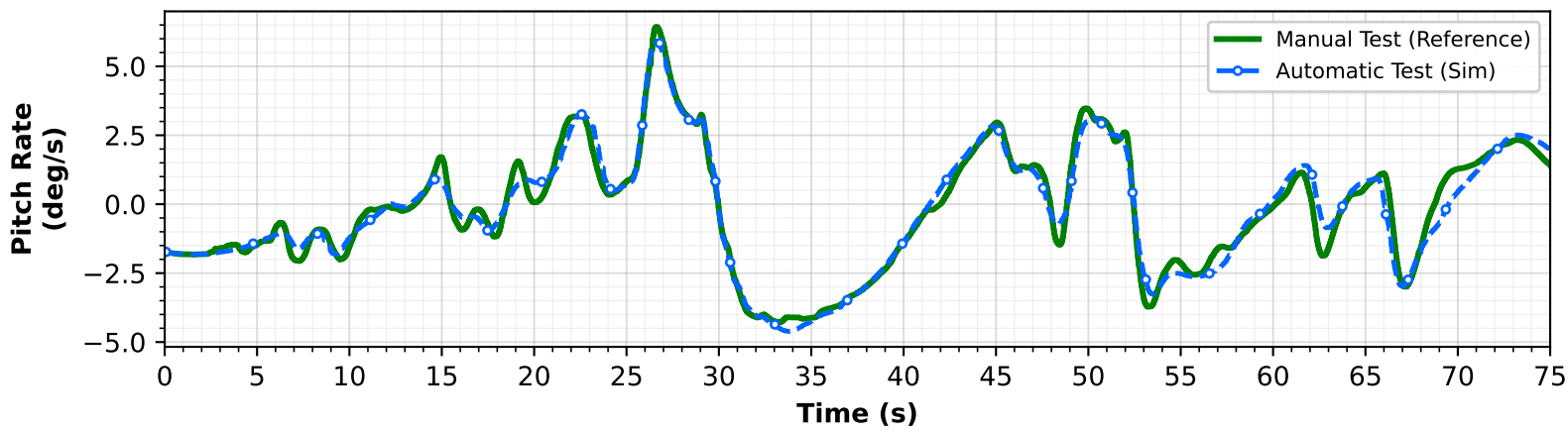

1st Quadrant Combined Group 1 (Airspeed, Climb, Altitude)
Indicated Airspeed (kts)
Manual Test (Reference)
Automatic Test (Sim)
POH Speed (80 kt)
Tolerance (±3 kt)
Time (s)
Rate of Climb/Descent (fpm)
Manual Test (Reference)
Automatic Test (Sim)
POH ROC (820 fpm)
Tolerance (±100 fpm)
Time (s)
Indicated Altitude (ft)
Manual Test (Reference)
Automatic Test (Sim)
Tolerance (-1000 ft)
Time (seconds)

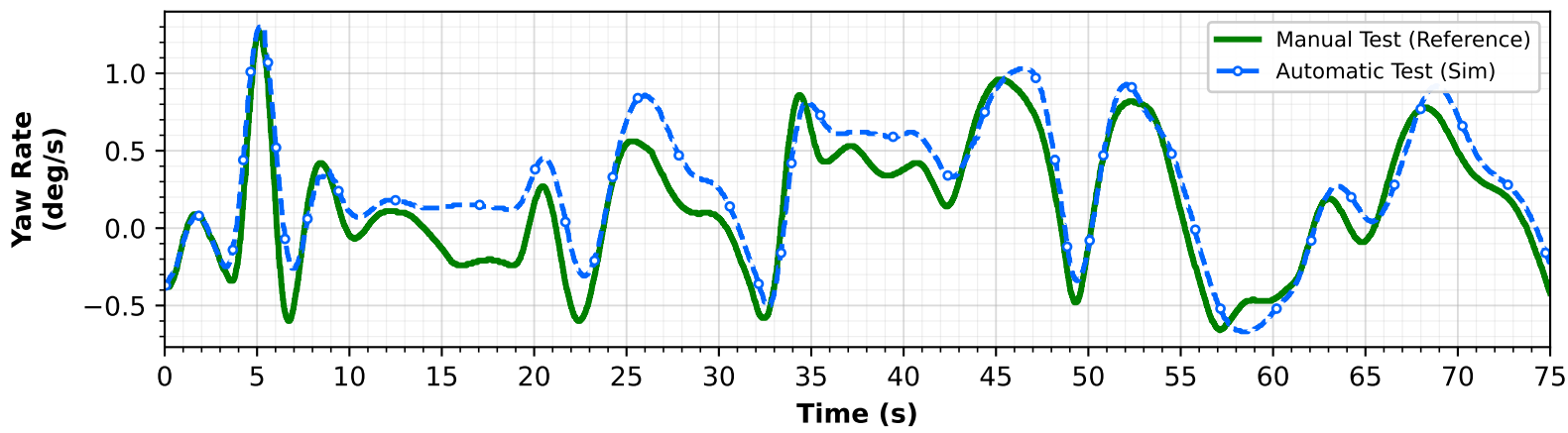
# 1st Quadrant Combined Group 2 (Pitch, Bank, Yaw, Heading)



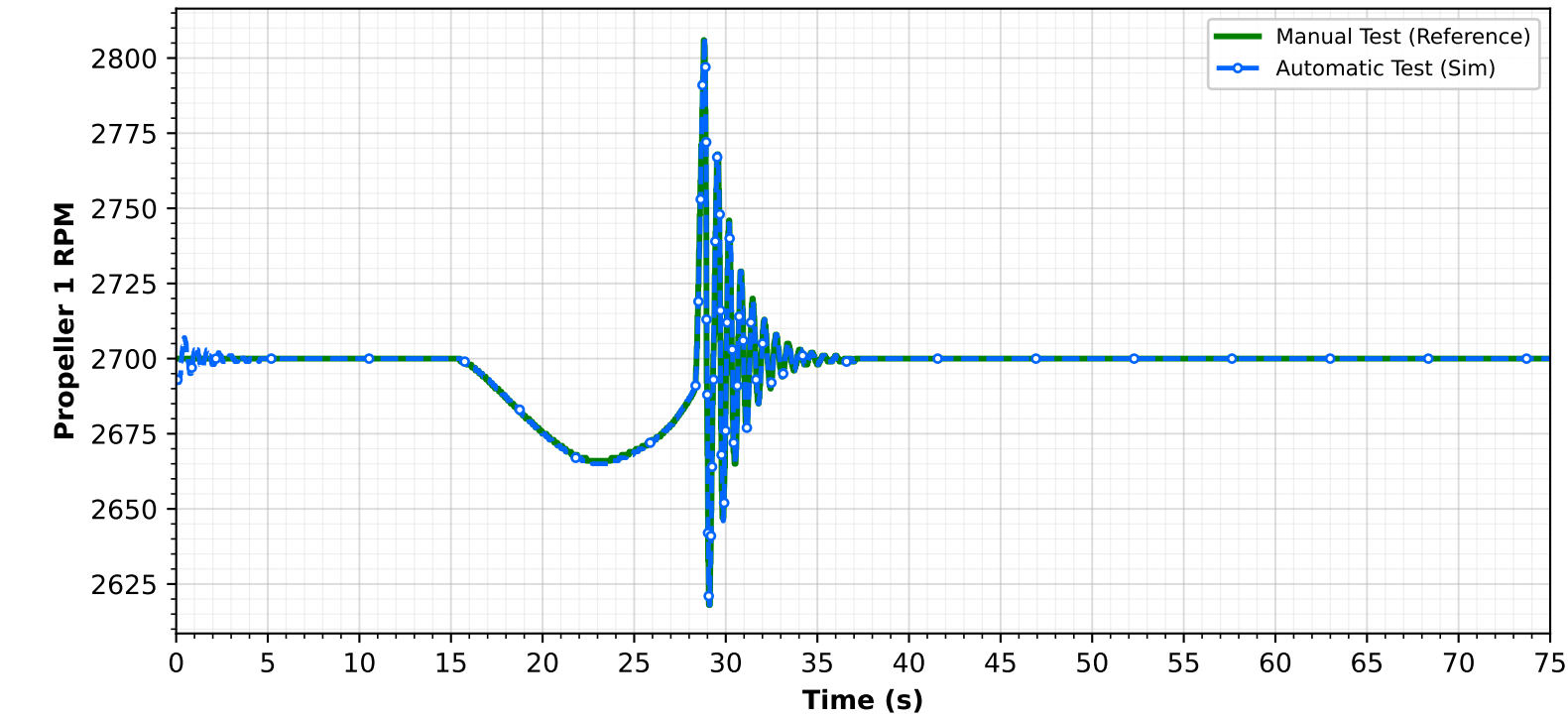



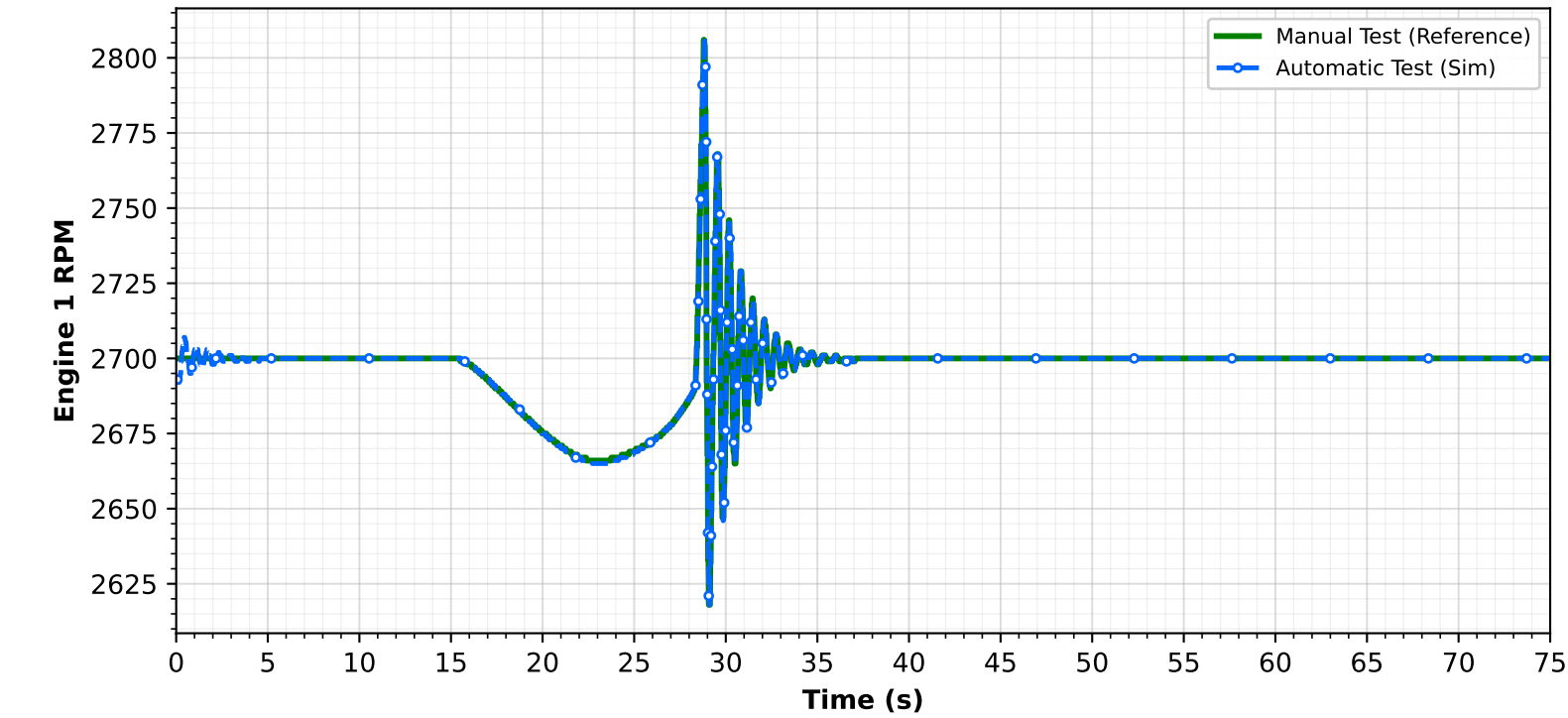



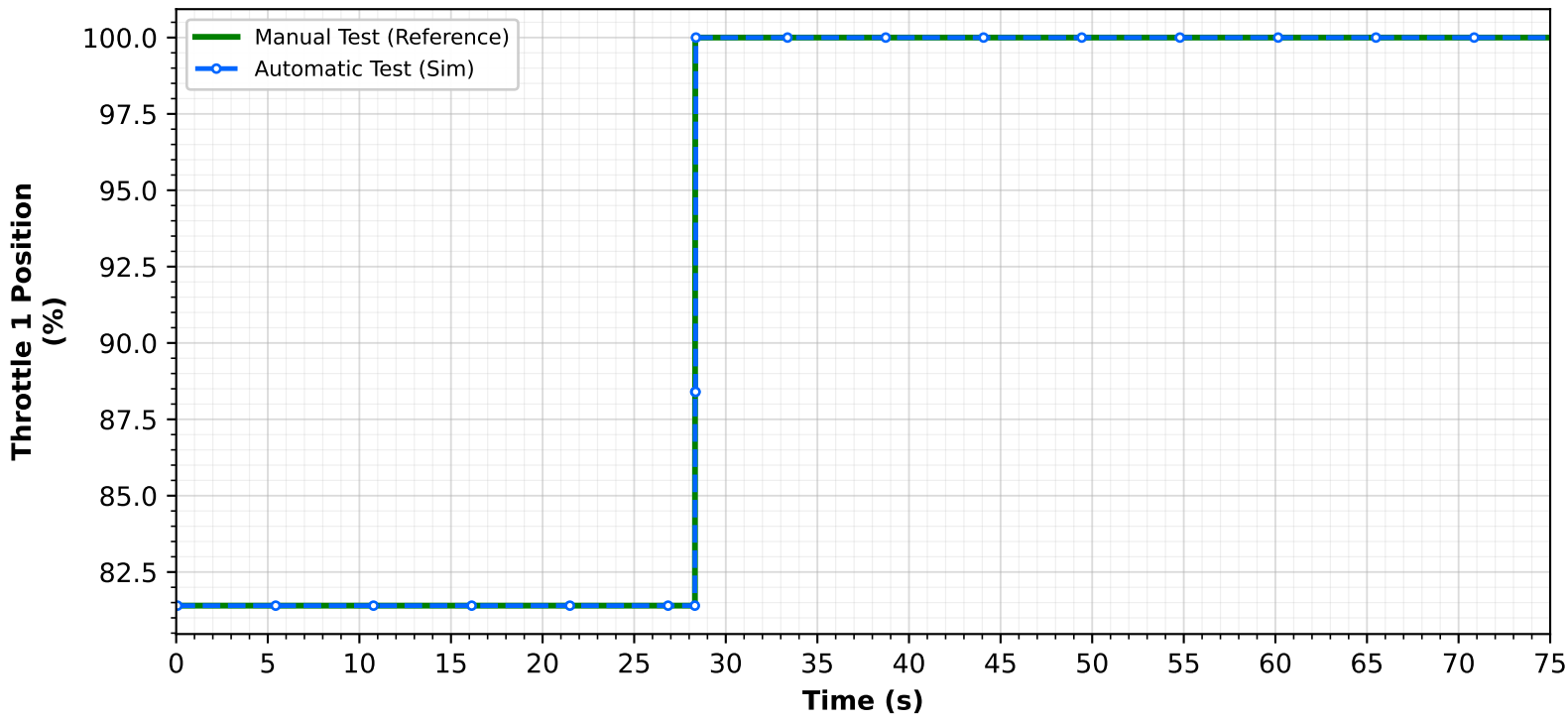



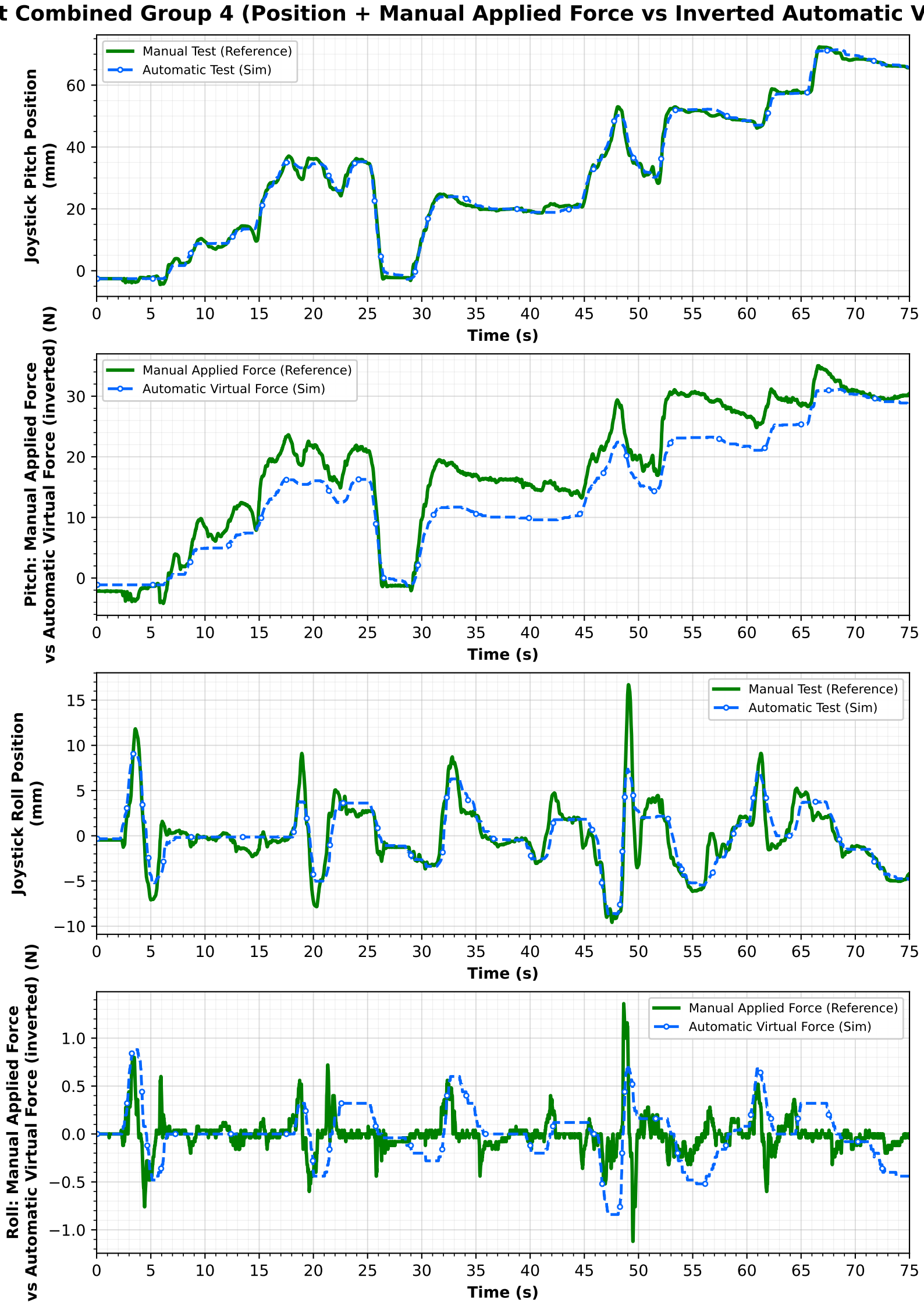



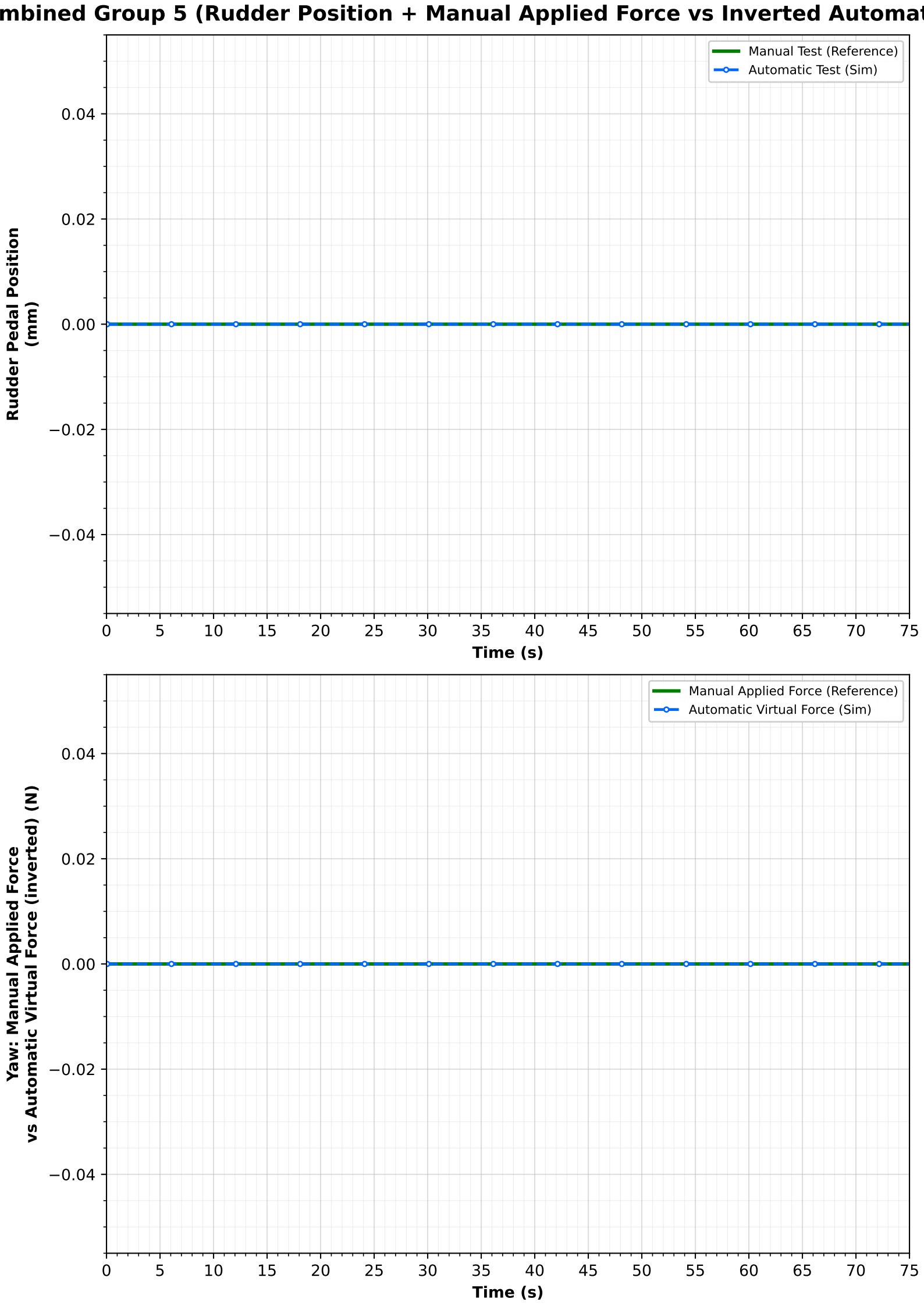
# 1st Quadrant Combined Group 3 (Propeller, Engine, Throttle)




t Combined Group 4 (Position + Manual Applied Force vs Inverted Automatic V
Manual Test (Reference)
Automatic Test (Sim)
Joystick Pitch Position (mm)
Time (s)
Manual Applied Force (Reference)
Automatic Virtual Force (Sim)
Pitch: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
Time (s)
Manual Test (Reference)
Automatic Test (Sim)
Joystick Roll Position (mm)
Time (s)
Manual Applied Force (Reference)
Automatic Virtual Force (Sim)
Roll: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
Time (s)


mbined Group 5 (Rudder Position + Manual Applied Force vs Inverted Automat
Manual Test (Reference)
Automatic Test (Sim)
Rudder Pedal Position (mm)
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
Time (s)
Manual Applied Force (Reference)
Automatic Virtual Force (Sim)
Yaw: Manual Applied Force vs Automatic Virtual Force (inverted) (N)
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
Time (s)